Notes on Continuous Time Model

$\boxed{\text{2nd Part}}$

Dynamic Macro II
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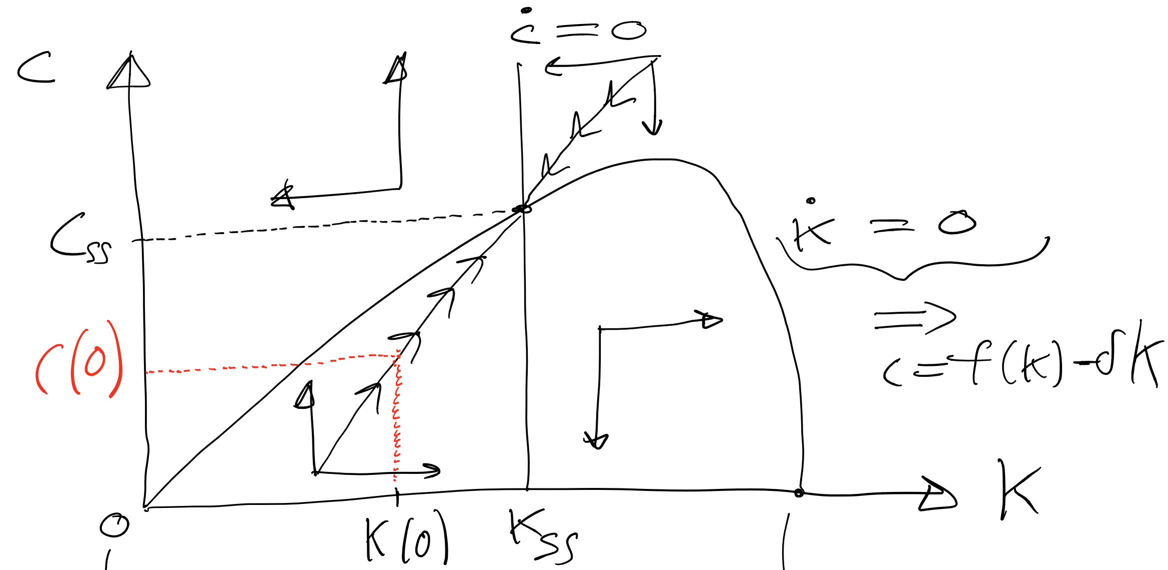
Let me draw again the phase diagram

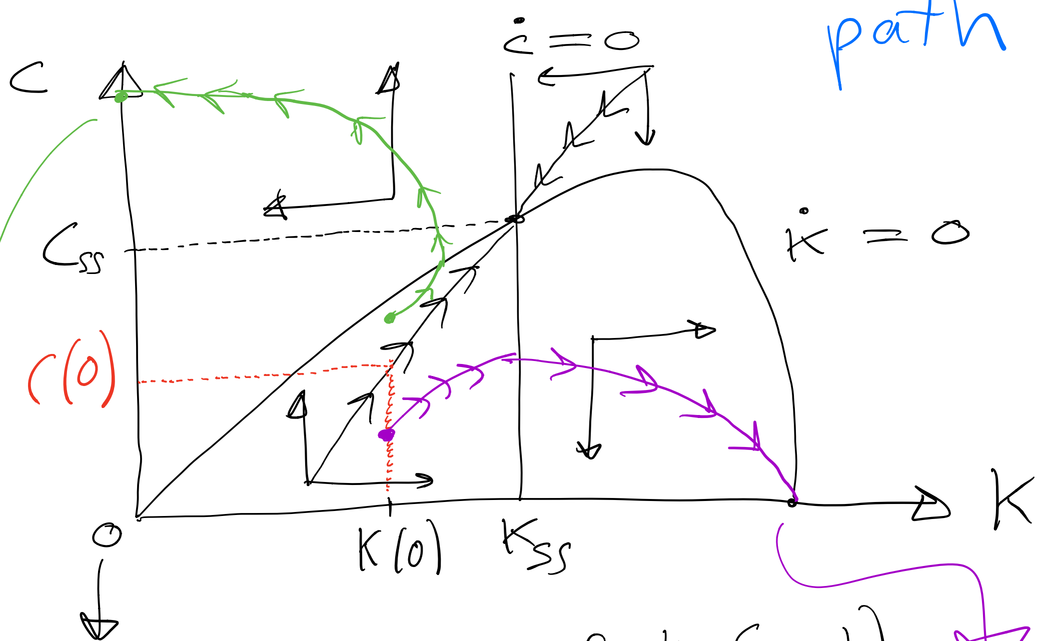


with $K=0$, $c=0$, from feasibility with $\dot{k}=0$

with sufficiently high K, the marginal increase in $f(k)$ is dominated by δk, reaching that point

What happens if we choose, or guess, a c(0) that is ~~not~~ on the path to the ss?

Put the point on the graph, let arrows determine the path



(Oops sorry for the Spanish)

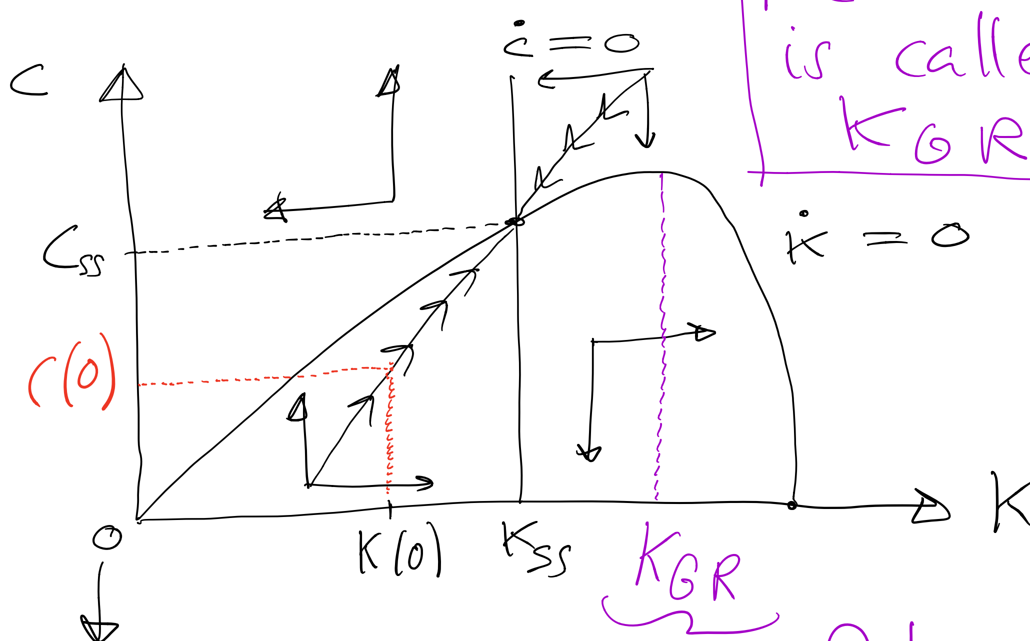
Al tocar eje de C, como K = 0, economía se colapsa, y salta al origen. Viola Euler, que describe comportamiento sin saltos

Acumulamos demasiado capital. Perdemos consumo. Viola Transversalidad

Final point: Golden Rule

It is easy to see that the maximum of $\{\dot{k} = 0 ;\ c = f(k) - \delta k\}$ is to the right of K_{ss}

$\boxed{\text{The maximizer is called } K_{GR}}$

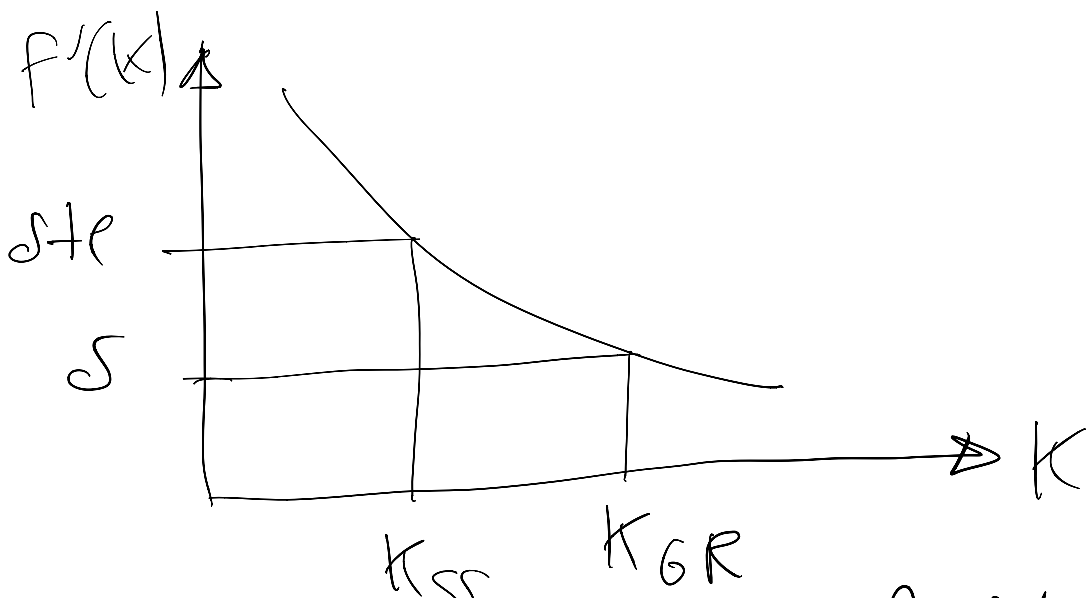


$\underbrace{K_{GR}}_{\text{Golden Rule capital}}$

K_{ss} comes from $f'(k) = \delta + \rho$

K_{GR} comes from $\left\{ \max\limits_{k}\ c = f(k) - \delta k \right\}$ with F.O.C. $f'(k) = \delta$

Given that $f'(k)$ is decreasing (because of decreasing marginal returns), k_{GR} will be larger



In the Planner's Problem, or in the Competitive Eq. (Welfare Theorems hold) the consumer does <u>not</u> reach k_{GR} because she is impatient:

$$\rho > 0$$